MFGOS PLATFORM ENGINEERING

16-Week Engineering Roadmap

Master technical tracking specification for the development lifecycle of **MFGOS (Configurable Manufacturing Operations System)**.

Document Version: 1.0.0 (GA Specification)
Platform: Web (React/Vite) + Mobile (Flutter)
Database: PostgreSQL / Supabase Core
Current State: Phase 1 Delivered & Operational

ROADMAP ARCHITECTURE & TIMELINE OVERVIEW

4 MONTHS • 16 SPRINTS

PHASE 1	PHASE 2	PHASE 3	PHASE 4
Weeks 01 – 04	**Weeks 05 – 08**	**Weeks 09 – 12**	**Weeks 13 – 16**
Foundation, Multi-Tenant Schema, Auth, Master Data & Stock Ledger	Manufacturing Core, Multi-Level BOM, Routing & Production Orders	Configurable Workflow FSM, Quality Gates & Flutter Offline Sync	Executive Analytics, AI Insights, Testing & Production Release
DELIVERED	SCHEDULED	SCHEDULED	SCHEDULED

Phase 1: Foundation, Security & Operational Master Data

WEEKS 01 – 04 • COMPLETED & OPERATIONAL

Week 01: System Specifications, Architecture & Project Scaffolding

Architecture & Foundation Setup

KEY ENGINEERING OBJECTIVES

- Establish Software Requirements Specification (SRS) following IEEE standards.
- Architect system topology: Monorepo structure, data isolation model, and API contracts.
- Formulate database ER diagram and relational constraints.
- Initialize Git repository with branch policies, `.gitignore`, and CI/CD foundations.

TECHNICAL DELIVERABLES

- `docs/srs.md`, `docs/architecture.md`, `docs/database-design.md`, `docs/testing-strategy.md`.
- Monorepo layout (`apps/web`, `apps/mobile`, `supabase`, `docs`).
- Configured GitHub remote at `https://github.com/VashimSipai/mfgos`.

CRISP WEEKLY SUMMARY

Week 1 finalized all core engineering documentation (IEEE-compliant SRS, System Architecture, Database Schema, and Testing Strategy), structured the monorepo workspace for Web and Mobile applications, and configured the Git repository with version control protocols.

Week 02: Multi-Tenant Schema, Supabase Auth & Role-Based Access Control (RBAC)

Multi-Tenant Identity & Access Management

KEY ENGINEERING OBJECTIVES

- Author and apply PostgreSQL migrations for organizations, plants, user profiles, roles, and permissions.
- Implement Row-Level Security (RLS) policies ensuring tenant and organization isolation.
- Setup authentication flow (Sign up, Sign in, Password recovery, Role attribution).
- Implement Super Admin, Production Manager, Warehouse Manager, and Operator role boundaries.

TECHNICAL DELIVERABLES

- `supabase/migrations/20261014000001_initial_schema.sql` (Tables, Foreign Keys, RLS policies).
- Web Authentication UI with responsive layout, session persistence, and role guards.
- Organization & Plant switcher in the navigation bar.

CRISP WEEKLY SUMMARY

Week 2 deployed the multi-tenant database schema with strict Row-Level Security policies, configured role-based access control across 7 manufacturing roles, and implemented the full authentication and organization switching system.

Week 03: Master Data Management (MDM) Engine

Foundational Manufacturing Catalog

KEY ENGINEERING OBJECTIVES

- Products and Raw Materials with SKU, specifications, and barcode generation.
- Units of Measure (UOM) with conversion factors (kg, g, pieces, meters, liters).
- Suppliers and Customers with lead times and contact directory.
- Warehouses, Aisles, and Bin locations; Work Centers and Standard Production Operations.

TECHNICAL DELIVERABLES

- Master Data UI modules in `apps/web/src/features/master-data/`.
- Client services, type-safe models, and validation schemas.
- Search, filtering, and pagination for items, suppliers, and locations.

CRISP WEEKLY SUMMARY

Week 3 engineered the Master Data Management engine, enabling full lifecycle administration of products, raw materials, units of measure, warehouses, work centers, suppliers, and standard operation definitions.

Week 04: Inventory Engine, Stock Movement Ledger & Central Dashboard

Transaction-Auditable Inventory Ledger

KEY ENGINEERING OBJECTIVES

- Implement immutable stock movement ledger: receipts, issues, transfers, and adjustments.
- Build database triggers / stored procedures enforcing non-negative inventory rules and atomic balance calculations.
- Implement stock receipts against suppliers and stock adjustments with audit remarks.
- Develop primary Management Operations Dashboard featuring key inventory metrics, quick actions, and warehouse capacity cards.

TECHNICAL DELIVERABLES

- `supabase/migrations/20261014000002_inventory_ledger_functions.sql`.
- Inventory management screens in `apps/web/src/features/inventory/`.
- Interactive Management Dashboard with rich UI aesthetics, KPIs, and responsive charts.
- Full demo seed data covering an engineering gear manufacturing plant.

CRISP WEEKLY SUMMARY

Week 4 delivered the transaction-safe inventory ledger with immutable stock movements, low-stock alerting, warehouse transfers, and the central Web Operations Dashboard equipped with real-time KPI metrics and seed data.

Phase 2: Manufacturing Core & Production Lifecycle

WEEKS 05 – 08 • SCHEDULED

Week 05: Bill of Materials (BOM) & Material Explosion Engine

Product Engineering & Recipe Management

KEY ENGINEERING OBJECTIVES

- Implement multi-level Bill of Materials (BOM) creation with revision tracking.
- Build component quantity calculators with scrap percentage allowance.
- Implement material explosion logic to calculate aggregate raw material needs for batch orders.
- Add BOM cost estimation based on standard material procurement prices.

TECHNICAL DELIVERABLES

- Interactive BOM Builder with drag-and-drop hierarchy.
- Material calculation API ensuring unit-of-measure conversion accuracy.

CRISP WEEKLY SUMMARY

Week 5 implemented the Bill of Materials (BOM) architecture with component revision control, scrap percentage estimation, and automated material requirement explosion for planned production runs.

Week 06: Manufacturing Routing & Work Center Scheduling

Production Process Definition

KEY ENGINEERING OBJECTIVES

- Define manufacturing routing sequences linking products to an ordered series of operations.
- Associate routing steps with target work centers, estimated cycle times, and setup durations.
- Implement work center capacity and bottleneck indicators.

TECHNICAL DELIVERABLES

- Routing creation and visual sequence editor.
- Work center assignment matrix.

CRISP WEEKLY SUMMARY

Week 6 created the manufacturing routing engine, mapping products to sequential operations, machine work centers, and estimated cycle times to enable accurate production planning.

Week 07: Production Order Lifecycle & Material Allocation

Shop Floor Order Generation

KEY ENGINEERING OBJECTIVES

- Build production order creation with target quantities, start/due dates, and batch identifiers.
- Automatic inventory check: highlight available, allocated, and shortage quantities.
- Implement transactional material reservation and material issuance to the shop floor.

TECHNICAL DELIVERABLES

- Production order management workbench with status lifecycles (`Draft` , `Planned` , `Released` , `In-Progress`).
- Shop floor material dispatch slips.

CRISP WEEKLY SUMMARY

Week 7 deployed the production order lifecycle manager, enabling automated material reservation, shortage detection, and material issue slips for release to the shop floor.

Week 08: Shop Floor Operation Execution & Scrap Tracking

Work-In-Progress (WIP) Tracking

KEY ENGINEERING OBJECTIVES

- Implement operator operation execution: Start, Pause, Resume, and Complete step.
- Record actual cycle times, operator identifiers, and completed vs. rejected counts.
- Automatic work-in-progress (WIP) transition to the next routing operation upon step completion.

TECHNICAL DELIVERABLES

- Shop floor job traveler interface.
- WIP tracking matrix and defect reason classification logging.

CRISP WEEKLY SUMMARY

Week 8 implemented shop-floor execution tracking for routing operations, capturing operator timestamps, real-time WIP transitions, and scrap quantity logging with defect categorization.

Phase 3: Configurability Engine & Mobile Offline Operations

WEEKS 09 – 12 • SCHEDULED

Week 09: Configurable Workflow State Machine

Dynamic Process Engine

KEY ENGINEERING OBJECTIVES

- Design the metadata-driven workflow engine storing stages, transition rules, and required roles in the database.
- Implement Administrative Visual Workflow Builder (customizing stages for Textile, Furniture, or Precision Engineering).
- Add transition validation: prevent unauthorized role jumps or skipping mandatory prerequisite stages.

TECHNICAL DELIVERABLES

- `workflows`, `workflow_stages`, and `workflow_transitions` database schema and API.
- Visual stage sequence builder with drag-and-drop reordering.

CRISP WEEKLY SUMMARY

Week 9 built the configurable workflow engine, enabling administrators to dynamically define arbitrary manufacturing stages, permitted transitions, and mandatory role permissions without code modifications.

Week 10: Multi-Stage Approvals & Quality Inspection Gates

Quality & Governance Checkpoints

KEY ENGINEERING OBJECTIVES

- Implement configurable Quality Inspection Templates (parameters, tolerances, pass/fail thresholds).
- Enforce gating transitions: a production stage cannot complete until the required Quality Inspector approves the inspection form.
- Implement digital sign-off and historical audit log of all transition events.

TECHNICAL DELIVERABLES

- Quality inspection template builder and execution modal.
- Audit log timeline showing "Who, What, When, and Outcome" for every order transition.

CRISP WEEKLY SUMMARY

Week 10 incorporated configurable quality inspection gates into the workflow engine, enforcing mandatory quality sign-offs and maintaining an immutable audit history of all stage transitions.

Week 11: Flutter Factory Terminal & Mobile Operations

Mobile Shop-Floor Client

KEY ENGINEERING OBJECTIVES

- Build responsive Flutter mobile application targeting factory floor operators and supervisors.
- Implement operator authentication, role-tailored daily task dashboard, and assigned job lists.
- Provide barcode/QR scanning to rapidly pull up production order travelers and raw material lot tags.

TECHNICAL DELIVERABLES

- Flutter application with clean Material 3 design in `apps/mobile/`.
- Operator task dispatch screen and mobile inspection forms.

CRISP WEEKLY SUMMARY

Week 11 developed the Flutter mobile factory terminal with Material 3 design, delivering operator task queues, step execution interfaces, and rapid barcode/QR scanning for shop-floor job travelers.

Week 12: Offline-First Architecture & Bidirectional Sync Engine

Industrial-Grade Offline Resilience

KEY ENGINEERING OBJECTIVES

- Integrate SQLite / Drift for local caching of assigned orders, BOMs, and inspection templates.
- Implement offline mutation queue: write operations locally with UUIDs and idempotency keys.
- Build background sync manager: auto-detect connectivity restoration, drain the sync queue, handle optimistic UI updates, and resolve concurrency conflicts.

TECHNICAL DELIVERABLES

- Local Drift database schema and repository layer in Flutter.
- Sync status indicator (Online / Offline / Syncing) with retry policies and duplicate protection.

CRISP WEEKLY SUMMARY

Week 12 engineered the offline-first SQLite/Drift synchronization engine for Flutter, enabling operators to record completions and inspections without connectivity and auto-reconcile with Supabase upon reconnection.

Phase 4: Analytics, AI Insights, Hardening & Deployment

WEEKS 13 – 16 • SCHEDULED

Week 13: Executive Analytics & Operational Reporting

Decision Intelligence & Metrics

KEY ENGINEERING OBJECTIVES

- Build interactive charts using Recharts for production throughput, Overall Equipment Effectiveness (OEE) approximation, and scrap rates.
- Generate exportable reports for inventory valuations, supplier receipt ledgers, and order completion histories.
- Implement customizable date-range filtering and plant comparison views.

TECHNICAL DELIVERABLES

- Executive Analytics dashboard in Web App.
- Report generator for management reviews.

CRISP WEEKLY SUMMARY

Week 13 constructed the executive analytics suite with dynamic charts for production throughput, scrap percentages, inventory turnover, and exportable compliance reports.

Week 14: AI Operational Insights & Anomaly Detection

Data-Backed Intelligence

KEY ENGINEERING OBJECTIVES

- Integrate LLM API to analyze structured operational records and generate grounded narrative summaries.
- Identify scrap anomalies (e.g., "Work Center CNC-02 scrap rate spiked 4.2% on Material Lot ST-402").
- Predict potential stockout dates based on current production order burn rates.

TECHNICAL DELIVERABLES

- AI Insights widget on the Web Dashboard with prompt grounding against actual PostgreSQL aggregates.
- Anomaly detection alerts for supervisors.

CRISP WEEKLY SUMMARY

Week 14 integrated the AI Operational Insights engine, providing automated, data-backed anomaly detection and scrap root-cause summaries without hallucinating metrics.

Week 15: End-to-End Testing, Security Auditing & Performance Optimization

System Verification & Validation

KEY ENGINEERING OBJECTIVES

- Execute automated integration tests for inventory consistency and negative stock prevention.
- Perform RLS security audit verifying organization isolation across all endpoints.
- Stress test offline queue with 50+ bulk offline transactions to verify conflict-free sync.
- Optimize database indexes and bundle size for lightning-fast loads.

TECHNICAL DELIVERABLES

- Automated test suites and test execution report.
- Security verification log documenting multi-tenant data isolation.

CRISP WEEKLY SUMMARY

Week 15 executed end-to-end testing, verifying atomic inventory transactions, multi-tenant RLS isolation, offline sync resilience under simulated network failure, and sub-second page performance.

Week 16: Deployment, Final Documentation & Production Release

Production Deployment & General Availability

KEY ENGINEERING OBJECTIVES

- Deploy Web Application to Vercel and Supabase cloud instance.
- Build Flutter production release bundle.
- Complete the comprehensive System Administration & Developer documentation.
- Prepare high-impact demonstration walkthrough video and operational scenario playbook.

TECHNICAL DELIVERABLES

- Live production web deployment link and downloadable Flutter release build.
- Technical Architecture Specification and API references.
- Dual-industry live demonstration scenario (Machining vs. Furniture).

CRISP WEEKLY SUMMARY

Week 16 finalized the production deployments, compiled the complete system documentation package, and verified the dual-industry operational demonstration workflow.

LIFECYCLE STATUS SUMMARY — **PHASE 1 OPERATIONAL**

PHASE	CORE FOCUS AREA	OPERATIONAL STATUS
Phase 1 (Weeks 01 – 04)	Foundation, Security, Master Data & Stock Movement Ledger	COMPLETED & OPERATIONAL
Phase 2 (Weeks 05 – 08)	Manufacturing Core, BOM, Routing & WIP Orders	SCHEDULED (Phase 2)
Phase 3 (Weeks 09 – 12)	Configurable Workflow Engine & Mobile Offline Sync	SCHEDULED (Phase 3)
Phase 4 (Weeks 13 – 16)	Analytics, AI Insights, Testing & Production Release	SCHEDULED (Phase 4)